

Evan Frangipane

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Education

Ph.D. in Physics *University of California, Santa Cruz*

Santa Cruz, CA *Aug 2018 - Aug 2024*

B.A. in Physics *University of California, Berkeley*

Berkeley, CA *Aug 2014 - Jun 2018*

Certifications

Google Advanced Data Analytics Professional Certificate

Feb 2025

Employment

Graduate Student Researcher, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Designed and executed Python-based data analysis pipelines to extract insights from large, complex datasets.
- Conducted hypothesis testing, causal inference analysis, and statistical modeling to support data-driven decision-making.

Teaching Assistant, *University of California, Santa Cruz*

Aug 2018 - Aug 2024

- Taught Python-based data analysis, covering statistical techniques, regression modeling, and machine learning applications.

Technical Projects

arxbot: Physics Chatbot via Rag Pipeline (Self-Directed)

July 2025

- Built a domain-specific chatbot capable of answering scientific queries by combining BM25 retrieval with local LLM generation via Ollama into a RAG pipeline.

LightPy: Simulation of Wave Properties of Light (Self-Directed)

July 2025

- Developed a modular, production-level Python project with unit tests and CI/CD automation using GitHub Actions, adhering to software engineering best practices.

End-to-End arXiv Abstract Classification (Self-Directed)

July 2025

- Developed and deployed a logistic regression model for real-time classification of arXiv abstracts, enabling live prediction serving in production.
- Experimented with fine-tuning DistilBERT using Hugging Face Transformers and LoRA (Low-Rank Adaptation), achieving improved accuracy with parameter-efficient training methods.

Multi-Model Fraud Detection: LR, SVM, and Random Forest (Self-directed)

Feb 2025

- Compared three models for credit card fraud detection on highly imbalanced (280k transactions, < 0.2% fraud) dataset.
- Random Forest outperformed logistic regression and SVM with an 0.84 AUCPR and 0.82 f1-score.

Multiple Testing Problem and the Bonferroni Correction (Self-directed)

Jan 2025

- Simulated the multiple testing problem in Python using Monte Carlo methods and applied the Bonferroni correction to control false positive rates.

Statistical Analysis of Primordial Black Hole Populations

Oct 2023

- Ran large-scale microlensing simulations on a distributed Kubernetes cluster and applied statistical tests to distinguish planet vs. black hole populations.

Skills

- **Programming & Tools:** Python (Pandas, scikit-learn, Pytorch), SQL, Git, Tableau, $\text{\LaTeX}$
- **Machine Learning & Statistics:** Supervised & Unsupervised Learning, Regression, A/B Testing, Hypothesis Testing, Time Series, Natural Language Processing, LLMs & Retrieval-Augmented Generation (RAG)
- **MLOps:** Model Deployment (FastAPI, Render, AWS Lambda), Containerization (Docker), Orchestration (Kubernetes), Experiment Tracking and Reproducibility (MLFlow)
- **Professional Skills:** Data-driven decision-making, stakeholder engagement, and impactful data storytelling